PADHAKU
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CHAPTER 9

Propositions and their Converses

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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🤔 Think and Reflect · Introduction

? **Statement 2: If two angles of a triangle are equal, then the sides opposite the equal angles have equal lengths. Is this statement true? Can you prove it? (Hint: Draw the altitude from the vertex containing the third angle.)**

✔ ANSWER

Yes, it is true. Proof:

1. In ΔABC, let $\angle B = \angle C$. Draw the altitude $AD \perp BC$.
2. In ΔABD and ΔACD: $\angle B = \angle C$ (given), $\angle ADB = \angle ADC = 90°$, $AD = AD$ (common).
3. So $\Delta ABD \cong \Delta ACD$ (AAS criterion).
4. Hence **AB = AC** (corresponding parts of congruent triangles).

So the sides opposite the equal angles are equal.

⭐ Examples

Ex 2 **Proposition P: If a number is a multiple of 6, then it is a multiple of 3. Justify why P is true. Is the converse Q (If a number is a multiple of 3, then it is a multiple of 6) true? If not, give a counterexample.**

✔ ANSWER

P: If n is a multiple of 6, then n = 6k = 3 × (2k), so n is a multiple of 3. **P is true.**

Q: **False.** Counterexample: 9 is a multiple of 3 (9 = 3 × 3) but not a multiple of 6.

So not all multiples of 3 are multiples of 6.

Ex 3 **n is a positive integer. P: If n is a perfect square, then it has an odd number of factors. Q: If n has an odd number of factors, then it is a perfect square. Which of them are true?**

✔ ANSWER

Both P and Q are true.

Factors come in factor–partner pairs whose product is n. Only a square n = f × f has a pair with the same number repeated, (f, f), and there can be only one such pair.

e.g. 36: (1, 36), (2, 18), (3, 12), (4, 9), (6, 6) → 9 factors (odd).

e.g. 12: (1, 12), (2, 6), (3, 4) → 6 factors (even).

Discussion **'If a number has an odd number of factors, there must be a factor-partner pair in which the same number repeats ... Thus, if a number has an odd number of factors, then it is a perfect square.' What does this argument prove? Statement P or Q?**

✔ ANSWER

It starts from ‘n has an odd number of factors’ and ends with ‘n is a perfect square’.

So it proves **Statement Q** (the converse), not P.

Ex 4 **P: If two triangles have the same area, then they are congruent. Q: If two triangles are congruent, then they have the same area. Determine if these are true. Justify the true statements and give a counterexample for each false one.**

✔ ANSWER

P is false. Counterexample: a triangle with base 6 cm and height 2 cm, and a triangle with base 4 cm and height 3 cm. Both have area ½ × 6 × 2 = ½ × 4 × 3 = 6 cm², but they are not congruent (different bases).

Q is true. Congruent triangles can be placed exactly on each other, so they cover the same region and have the same area.

? **It can also happen that both the proposition and its converse are false! Can you give an example? (Think and Reflect)**

💡 SAMPLE ANSWER

P: If a number is a multiple of 2, then it is a multiple of 3. **False** — 4 is a multiple of 2 but not of 3.

Converse: If a number is a multiple of 3, then it is a multiple of 2. **False** — 9 is a multiple of 3 but not of 2.

Another: 'If a quadrilateral is a rectangle, then it is a rhombus' and its converse are both false (a 4 cm × 2 cm rectangle; a rhombus with angles 60° and 120°).

Ex 5 **Converse of the Baudhāyana–Pythagoras theorem: Let a, b, c be the sidelengths of a triangle. If $a^2 + b^2 = c^2$, then the triangle is right-angled. ΔABC has sides a, b, c with $a^2 + b^2 = c^2$. ΔXYZ is a right triangle (∠Z = 90°) with YZ = a and XZ = b. What can we say about the length of XY? What can we say about ΔABC and ΔXYZ?**

✔ ANSWER

In right ΔXYZ, by the Baudhāyana–Pythagoras theorem: $XY^2 = a^2 + b^2 = c^2$.

So **XY = c**.

Now BC = YZ = a, CA = ZX = b, AB = XY = c.

So **ΔABC ≅ ΔXYZ** (SSS criterion).

Hence ∠C = ∠Z = 90°, so ΔABC is right-angled. The converse is true.

🤔 Think and Reflect · A Note

1 **Identify real-life examples in which a proposition is true but not the converse. Give**

nice counterexamples!

💡 SAMPLE ANSWER

- P: If an animal is a cow, then it has four legs (true). Converse: If an animal has four legs, then it is a cow — false (a dog has four legs).
- P: If a person lives in Delhi, then the person lives in India (true). Converse false: a person living in Mumbai lives in India but not in Delhi.
- P: If it is a Sunday, then the school is closed (true). Converse false: the school is also closed on holidays like 15 August.

2 **Give more examples from geometry as well as number theory in which a proposition is true but not its converse. Give appropriate counterexamples.**

💡 SAMPLE ANSWER

Geometry:

- If a quadrilateral is a rhombus, then its diagonals are perpendicular (true). Converse false: a kite has perpendicular diagonals but is not always a rhombus.
- If a quadrilateral is a rectangle, then its diagonals are equal (true). Converse false: an isosceles trapezium has equal diagonals.

Number theory:

- If n is divisible by 4, then n is even (true). Converse false: 6 is even but not divisible by 4.
- If n is a prime greater than 2, then n is odd (true). Converse false: 9 is odd but not prime.

📝 Exercise Set 9.1

1 **If two lines are parallel, then the corresponding angles formed by a transversal are equal.**

✔ ANSWER

Converse: If the corresponding angles formed by a transversal of two lines are equal, then the lines are parallel.

Proposition: **True** — this is the property of parallel lines cut by a transversal.

Converse: **True** — if the lines were not parallel, they would meet and form a triangle with the transversal; then one corresponding angle would be an exterior angle equal to an interior opposite angle, which is impossible (an exterior angle = sum of the two interior opposite angles, so it is larger). So the lines are parallel.

2 **If a quadrilateral is a square, then all its angles are equal.**

✔ ANSWER

Converse: If all the angles of a quadrilateral are equal, then it is a square.

Proposition: **True** — every angle of a square is 90°.

Converse: **False.** Counterexample: a rectangle with sides 4 cm and 2 cm has all angles 90° but is not a square.

3* **Given any ΔABC, the bisectors of ∠B and ∠C meet at the incentre I and are extended to meet the opposite sides at E (on AC) and F (on AB) (Fig. 9.1).**

Proposition: If AB = AC, then IE = IF.

💡 SAMPLE ANSWER

Converse: If IE = IF, then AB = AC.

Proposition: True. Proof:

1. AB = AC ⇒ ∠B = ∠C ⇒ ½∠B = ½∠C, i.e. ∠IBC = ∠ICB and ∠IBF = ∠ICE.

2. From ∠IBC = ∠ICB, IB = IC.

3. In ΔIBF and ΔICE: ∠IBF = ∠ICE, IB = IC, ∠BIF = ∠CIE (vertically opposite).

4. So ΔIBF ≅ ΔICE (ASA), hence **IF = IE**.

Converse: False. Counterexample: take ∠A = 60°, ∠B = 80°, ∠C = 40°. Then AB ≠ AC (∠B ≠ ∠C), but IE = IF.

Why IE = IF here: ∠IBC + ∠ICB = ½(80° + 40°) = 60°, so ∠BIC = 120°, and ∠FIE = ∠BIC = 120° (vertically opposite).

In quadrilateral AFIE, $\angle A + \angle FIE = 60° + 120° = 180°$, so AFIE is a cyclic quadrilateral (opposite angles add up to 180°).

AI bisects $\angle A$, so $\angle FAI = \angle EAI = 30°$. In a circle, equal angles at the circumference stand on equal chords, so **IF = IE**.

So IE = IF although AB ≠ AC: the converse is false. (The same works in every triangle with $\angle A = 60°$.)

4 **If x = y, then a + x = a + y, where x, y and a are any three numbers.**

✔ ANSWER

Converse: If a + x = a + y, then x = y.

Proposition: **True** — adding the same number a to equal numbers gives equal results.

Converse: **True** — subtract a from both sides: (a + x) − a = (a + y) − a, so x = y.

5 **If a and b are perfect squares, then ab is a perfect square.**

✔ ANSWER

Converse: If ab is a perfect square, then a and b are perfect squares.

Proposition: **True** — if $a = p^2$ and $b = q^2$, then $ab = p^2q^2 = (pq)^2$.

Converse: **False.** Counterexample: a = 2, b = 8. ab = 16 = 4^2 is a perfect square, but 2 and 8 are not.

6 **x, y are real numbers. If x = y, then $x^2 = y^2$.**

✔ ANSWER

Converse: If $x^2 = y^2$, then x = y.

Proposition: **True** — $x^2 = x \times x = y \times y = y^2$.

Converse: **False.** Counterexample: x = 2, y = −2. $x^2 = 4 = y^2$, but $x \neq y$.

7 **x, y are real numbers. If x = y, then $x^3 = y^3$.**

✔ ANSWER

Converse: If $x^3 = y^3$, then x = y.

Proposition: **True** — $x^3 = x \times x \times x = y \times y \times y = y^3$.

Converse: **True.** $x^3 - y^3 = (x - y)(x^2 + xy + y^2) = 0$.

$x^2 + xy + y^2 = (x + y/2)^2 + 3y^2/4$, which is 0 only when x = y = 0.

So either x − y = 0 or x = y = 0; in both cases **x = y**.

8 **n is a positive integer. If n is divisible by 24, then it is divisible by both 4 and 6.**

✔ ANSWER

Converse: If n is divisible by both 4 and 6, then it is divisible by 24.

Proposition: **True** — $n = 24k = 4 \times (6k) = 6 \times (4k)$.

Converse: **False.** Counterexample: 12 is divisible by 4 and by 6, but not by 24.

9 **n is a positive integer. If n is divisible by 60, then it is divisible by both 5 and 12.**

✔ ANSWER

Converse: If n is divisible by both 5 and 12, then it is divisible by 60.

Proposition: **True** — $n = 60k = 5 \times (12k) = 12 \times (5k)$.

Converse: **True.** $n = 12k$. 5 divides $12k$, and 5 is a prime that does not divide 12, so 5 divides k: $k = 5m$.

Then $n = 12 \times 5m = 60m$, so n is divisible by 60. (5 and 12 have no common factor other than 1.)

10 **n is a positive integer. If n is the square of a prime number, then it has exactly 3 factors.**

✔ ANSWER

Converse: If n has exactly 3 factors, then n is the square of a prime number.

Proposition: **True** — if $n = p^2$ (p prime), its factors are only 1, p, p^2 → exactly 3.

Converse: **True.** n has an odd number of factors, so n is a perfect square, $n = m^2$ ($m > 1$).

Then 1, m, m^2 are already 3 factors. If m had a factor d with $1 < d < m$, d would be a 4th factor.

So m has no such factor, i.e. m is prime and **$n = (\text{prime})^2$**.

11 **n is a positive integer. If n is a product of two unequal prime numbers, then it has exactly 4 divisors.**

✔ ANSWER

Converse: If n has exactly 4 divisors, then n is a product of two unequal prime numbers.

Proposition: **True** — if $n = pq$ ($p \neq q$ primes), its divisors are only 1, p, q, pq → exactly 4.

Converse: **False.** Counterexample: 8 has exactly 4 divisors (1, 2, 4, 8), but $8 = 2^3$ is not a product of two unequal primes.

12 **n is a positive integer. If n and n + 3 have no factors in common, then n is not a multiple of 3.**

✔ ANSWER

Converse: If n is not a multiple of 3, then n and n + 3 have no factors in common (other than 1).

Proposition: **True** — if n were a multiple of 3, then n + 3 would also be a multiple of 3, so 3 would be a common factor.

Converse: **True** — any common factor d of n and n + 3 divides (n + 3) − n = 3, so d = 1 or 3. d = 3 is not possible since n is not a multiple of 3. So d = 1.

e.g. n = 4: 4 and 7 have no common factor.

13 (i) **Find a counterexample: All numbers of the form $4n^2 + 1$ are prime.**

✔ ANSWER

n = 1: 5, n = 2: 17, n = 3: 37 (prime), but

n = 4: $4 \times 16 + 1 = 65 = 5 \times 13$.

65 is not prime.

13 (ii) **Find a counterexample: All numbers of the form $n^2 + n + 11$ are prime.**

✔ ANSWER

n = 10: $100 + 10 + 11 = 121 = 11 \times 11$.

121 is not prime. (Also n = 11 gives $143 = 11 \times 13$.)

13 (iii) **Find a counterexample: All numbers of the form $4^n + 3$ are prime.**

✔ ANSWER

n = 1: 7, n = 2: 19, n = 3: 67 (prime), but

n = 4: $256 + 3 = 259 = 7 \times 37$.

259 is not prime. (If n = 0 is allowed: $4^0 + 3 = 4$, also not prime.)

14 (i) **If n is a prime number, then $2^n - 1$ is a prime number.**

✔ ANSWER

n = 2: 3, n = 3: 7, n = 5: 31, n = 7: 127 (prime), but

n = 11: $2^{11} - 1 = 2048 - 1 = 2047 = 23 \times 89$.

11 is prime but 2047 is not.

14 (ii) **If n is an even number, then $2^n + 1$ is a prime number.**

✔ ANSWER

n = 2: 5, n = 4: 17 (prime), but

n = 6: $2^6 + 1 = 65 = 5 \times 13$.

6 is even but 65 is not prime.

15 (i) **'If a number is divisible by 8, then it is divisible by both 2 and 4.' Justify the statement.**

✔ ANSWER

If n is divisible by 8, then n = 8k.

n = 2 × (4k), so n is divisible by 2.

n = 4 × (2k), so n is divisible by 4.

So the statement is true.

15 (ii) **To check whether a given number is divisible by 8, is it enough to check whether it is divisible by 2 and 4? Why or why not?**

✔ ANSWER

No. That would be the converse, which is false.

Counterexample: 12 is divisible by 2 and by 4, but not by 8 (also 4, 20, 28 ...).

A number divisible by 4 is already divisible by 2, so 'divisible by 2 and 4' only means 'divisible by 4'.

Use the shortcut for 8: check if the number formed by the last three digits is divisible by 8.

16 **Express the relationship between 'a number is divisible by 3' and 'sum of the digits is a multiple of 3' using 'If-then' sentences.**

✔ ANSWER

- **If a number is divisible by 3, then the sum of its digits is a multiple of 3.**
- **If the sum of the digits of a number is a multiple of 3, then the number is divisible by 3.**

Both statements (a proposition and its converse) are true, e.g. 471: 4 + 7 + 1 = 12, and 471 = 3 × 157; 472: 4 + 7 + 2 = 13, and 472 is not divisible by 3.

17 (i) **A quadrilateral of type Q is made by putting two thin sticks together as diagonals and joining their endpoints (Fig. 9.2).**

Q satisfies: If a quadrilateral is of type Q, then it has equal-length diagonals.

(a) Should the two sticks be of equal length? Why or why not?

(b) Will it matter how the two sticks are put together?

✔ ANSWER

(a) **Yes.** Every quadrilateral of type Q has equal diagonals, so with sticks of unequal length we can never get type Q.

(b) **It may matter.** The property only says equal diagonals are needed; it does not say equal diagonals are enough. Type Q may need more (e.g. if Q were 'square', the sticks must also bisect each other at right angles). So equal sticks put together in any way may not give type Q.

17 (ii) **Instead, suppose Q satisfies: If a quadrilateral has equal diagonals, then it is of type Q. What will be your answers to (a) and (b) now?**

✔ ANSWER

(a) Equal sticks are **enough**: any quadrilateral with equal diagonals is of type Q. But they need not be equal — the property does not say every type-Q quadrilateral has equal diagonals, so some type-Q quadrilaterals may have unequal diagonals. Using equal sticks is the sure way.

(b) **No, it does not matter** how the equal sticks are put together (as long as they cross and the joined endpoints form a quadrilateral): the diagonals are equal, so the quadrilateral is of type Q.

🎉 **Chapter 9 done! Revise the answers once more tomorrow.**